(3) $\sigma = \frac{1}{3}\alpha V^{1/4} U^{3/4}$

$$\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_V = \alpha V^{1/4} U^{-1/4} \Rightarrow U^{1/4} = \alpha V^{1/4} \tau$$

$$U = \alpha^4 V \tau^4$$

$$P = \tau \left(\frac{\partial \sigma}{\partial V}\right)_U = \frac{1}{3}\alpha V^{-3/4} U^{3/4} \tau = \frac{1}{3}\alpha V^{-3/4}\left(\alpha^3 V^{\frac{1}{4}} \tau\right)^3 \tau =$$

$$= \frac{1}{3}\alpha^4 \tau^4$$

$$\Rightarrow PV = \frac{1}{3}\alpha^4 \tau^4 V = \frac{1}{3} U$$

(1)
g_1 — 2ε
2 — ε
1 — 0

(a) $Z_1 = \sum_n e^{-\varepsilon_n/\tau} = 1 \cdot e^0 + 2 \cdot e^{-\varepsilon/\tau} + 1 \cdot e^{-2\varepsilon/\tau} = 1 + 2e^{-\varepsilon/\tau} + e^{-2\varepsilon/\tau} =$

$$= \left(1 + e^{-\varepsilon/\tau}\right)^2$$

(b) $Z_N = Z_1^N = \left(1 + e^{-\varepsilon/\tau}\right)^{2N}$ (particles are distinguishable by their positions)

(c) $F_N = -\tau \ln Z_N = -2N\tau \ln\left(1 + e^{-\varepsilon/\tau}\right)$

$$\sigma_N = -\left(\frac{\partial F_N}{\partial \tau}\right)_V = 2N \ln\left(1 + e^{-\varepsilon/\tau}\right) + 2N\tau \frac{\frac{\varepsilon}{\tau^2} e^{-\varepsilon/\tau}}{1 + e^{-\varepsilon/\tau}} =$$

$$= 2N \ln\left(1 + e^{-\varepsilon/\tau}\right) + 2N \frac{\varepsilon/\tau}{e^{\varepsilon/\tau} + 1}$$

$$F_N = U_N - \tau\sigma_N \Rightarrow U_N = F_N + \tau\sigma_N = 2N \frac{\varepsilon}{\left(e^{\varepsilon/\tau} + 1\right)}$$

(d) $$C_V = \left(\frac{\partial U}{\partial \tau}\right)_V = 2N \frac{\partial}{\partial \tau} \frac{\varepsilon}{e^{\varepsilon/\tau}+1} = 2N\varepsilon \frac{-1\cdot e^{\varepsilon/\tau}}{\left(e^{\varepsilon/\tau}+1\right)^2}\left(-\frac{\varepsilon}{\tau^2}\right)$$

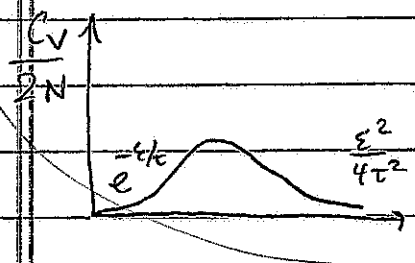
$$= 2N \frac{\varepsilon^2 e^{\varepsilon/\tau}}{\tau^2\left(1+e^{\varepsilon/\tau}\right)^2}$$

<u>Limits:</u> $\underline{\tau \gg \varepsilon} \Rightarrow \frac{\varepsilon}{\tau} \ll 1 \Rightarrow e^{\varepsilon/\tau} \cong 1 + \frac{\varepsilon}{\tau}$

$$C_V \approx 2N \frac{\varepsilon^2}{\tau^2 \cdot 4} = \frac{N}{2} \frac{\varepsilon^2}{\tau^2}$$

$\underline{\tau \ll \varepsilon} \Rightarrow \frac{\varepsilon}{\tau} \gg 1 \Rightarrow e^{\varepsilon/\tau} \gg 1$

$$C_V \approx 2N \frac{\varepsilon^2}{\tau^2} \frac{e^{\varepsilon/\tau}}{e^{2\varepsilon/\tau}} = 2N \frac{\varepsilon^2}{\tau^2} e^{-\varepsilon/\tau}$$



(2) Two-dimensional gas $\Rightarrow$ $\varepsilon(\vec{p}) = \dfrac{p_x^2 + p_y^2}{2m}$

(a) $Z_N = \dfrac{(Z_1)^N}{N!}$ for N indistinguishable particles.

$$Z_1 = \int \frac{dx\,dy\,dp_x\,dp_y}{(2\pi\hbar)^2}\, e^{-\frac{p_x^2+p_y^2}{2m\tau}} = \frac{V}{(2\pi\hbar)^2}\left(\int_{-\infty}^{\infty} dp\, e^{-\frac{p^2}{2m\tau}}\right)^2 =$$

$(p = z\sqrt{2m\tau})$

$$= \frac{V}{(2\pi\hbar)^2}\left(\sqrt{2m\tau}\int_{-\infty}^{\infty} dz\, e^{-z^2}\right)^2 = \frac{V}{(2\pi\hbar)^2}\, 2m\tau\cdot\pi = \frac{L^2 m\tau}{2\pi\hbar^2}$$

$$\Rightarrow Z_N = \frac{V^N}{N!}\left(\frac{2\pi m\tau}{(2\pi\hbar)^2}\right)^N, \quad F = -\tau \ln Z$$

(b) $U_N = \tau^2 \dfrac{\partial}{\partial\tau} \ln Z_N = \tau^2 N \dfrac{\partial}{\partial\tau} \ln\left(\dfrac{2\pi m\tau}{(2\pi\hbar)^2}\right) = N\tau$

(c) $$\sigma = -\left(\frac{\partial F}{\partial\tau}\right)_{V,N} = \frac{\partial}{\partial\tau}\left(\tau \ln Z\right)_{V,N} = \ln Z + \tau\frac{N}{\tau} =$$

$$= N \ln \frac{V\, 2\pi m\tau}{(2\pi\hbar)^2} - \ln N! + N = N \ln\left[\frac{V}{N}\,\frac{2\pi m\tau}{(2\pi\hbar)^2}\right] + 2N =$$

$$\ln N! = N \ln\frac{N}{e} = N\ln N - N \qquad = N\left\{\ln\left(\frac{V}{N}\,\frac{2\pi m\tau}{(2\pi\hbar)^2}\right) + 2\right\}.$$

(d) $$\mu = \left(\frac{\partial F}{\partial N}\right)_{\tau,V} = -\tau\frac{\partial}{\partial N}\left(\ln Z_N\right)_{\tau,V} =$$

$$= -\tau\frac{\partial}{\partial N}\left(N\ln\frac{V\, 2\pi m\tau}{(2\pi\hbar)^2} - N\ln\frac{N}{e}\right) = -\tau\left(\ln\frac{V\, 2\pi m\tau}{(2\pi\hbar)^2} - \ln N\right) =$$

$$= -\tau \ln\left[\frac{V}{N}\,\frac{2\pi m\tau}{(2\pi\hbar)^2}\right].$$

(4) Harmonic oscillator $\varepsilon_n = \left(n+\frac{1}{2}\right)\hbar\omega$

$$Z = \sum_n e^{-\varepsilon_n/\tau} = \sum_{n=0}^{\infty} e^{-\left(n+\frac{1}{2}\right)\hbar\omega/\tau} = e^{-\frac{\hbar\omega}{2\tau}} \sum_{n=0}^{\infty} e^{-\frac{\hbar\omega}{\tau}\cdot n} =$$

$$= \frac{e^{-\frac{\hbar\omega}{2\tau}}}{1-e^{-\hbar\omega/\tau}}$$

Probability to be in state k : $p_k = \dfrac{e^{-\varepsilon_k/\tau}}{Z}$

Probability to be in odd-k state: $P_{odd} = \sum_{k=odd} p_k$

odd: $k = 2l+1$, $l = 0, 1, 2 \ldots$

$$P_{odd} = \frac{1}{Z}\sum_{l=0}^{\infty} e^{-\left(2l+1+\frac{1}{2}\right)\frac{\hbar\omega}{\tau}} = \frac{e^{-\frac{3\hbar\omega}{\tau}}}{Z}\sum_{l=0}^{\infty} e^{-\frac{\hbar\omega}{\tau}\cdot 2l} =$$

$$= \frac{e^{-\frac{3\hbar\omega}{\tau}}}{Z}\left\{1 + e^{-\frac{2\hbar\omega}{\tau}} + \left(e^{-\frac{2\hbar\omega}{\tau}}\right)^2 + \ldots\right\} =$$

$$= \frac{e^{-3\hbar\omega/\tau}/Z}{1-e^{-\frac{2\hbar\omega}{\tau}}} = \frac{e^{-\frac{3\hbar\omega}{\tau}}}{1-e^{-\frac{2\hbar\omega}{\tau}}} \cdot \frac{1-e^{-\frac{\hbar\omega}{\tau}}}{e^{-\frac{\hbar\omega}{2\tau}}} =$$

$$= \frac{e^{-\frac{\hbar\omega}{\tau}}\left(1-e^{-\frac{\hbar\omega}{\tau}}\right)}{\left(1-e^{-\frac{\hbar\omega}{\tau}}\right)\left(1+e^{-\frac{\hbar\omega}{\tau}}\right)} = \frac{e^{-\hbar\omega/\tau}}{1+e^{-\hbar\omega/\tau}} = \frac{1}{e^{\frac{\hbar\omega}{\tau}}+1}$$

<u>limits</u> $P_{odd} = \begin{cases} e^{-\hbar\omega/\tau}, & \tau \to 0 \\ \frac{1}{2}, & \tau \to \infty \end{cases}$

because SHO in in the ground state $\underline{n=0}$ which is <u>even</u>

(equally likely to be in odd/even)